gravitational interaction. Each such possibility suggests a particular kind of Hamiltonian. And if you know the Hamiltonian, you know everything; it's a compact way of capturing all the dynamics of a physical system.

If a quantum wave function describes a system with some definite value of the energy, the Hamiltonian simply equals that value, and the Schrödinger equation implies that the system just keeps doing the same thing, maintaining a fixed energy. More often, since wave functions are superpositions of different possibilities, the system will be a combination of multiple energies. In that case the Hamiltonian captures a bit of all of them. The bottom line is that the right-hand side of Schrödinger's equation is a way of characterizing how much energy is carried by each of the contributions to a wave function in a quantum superposition; high-energy components evolve quickly, low-energy ones evolve more slowly.

What really matters is that there is some specific deterministic equation. Once you have that, the world is your playground.

Wave mechanics made a huge splash, and before too long Schrödinger, English physicist Paul Dirac, and others demonstrated that it was essentially equivalent to matrix mechanics, leaving us with a unified theory of the quantum world. Still, all was not peaches and cream. Physicists were left with the question that we are still struggling with today: What *is* the wave function, really? What physical thing does it represent, if any?

In de Broglie's view, his matter waves served to guide particles around, not to replace them entirely. (He later developed this idea into pilot-wave theory, which remains a viable approach to quantum foundations today, although it is not popular among working physicists.) Schrödinger, by contrast, wanted to do away with fundamental particles entirely. His original hope was that his equation would describe localized packets of vibrations, confined to a relatively small region of space, so that each packet would appear particle-like to a macroscopic observer. The wave function could be thought of as representing the density of mass in space.

Alas, Schrödinger's aspirations were undone by his own equation. If we start with a wave function describing a single particle approximately localized in some empty region of space, the Schrödinger equation is clear